

Shu Quan

quanshu@stu.pku.edu.cn | quanshu01.github.io | Google Scholar | GitHub

Education

Peking University

Beijing, China

M.S. in Environmental Science, GPA: 3.61/4.0

Sep. 2024 – Jun. 2027 (Expected)

- Research focus: AI safety and LLM evaluation with PKU-Alignment Group; advisor: Prof. Yaodong Yang.
- Visiting research: LLM hallucination mitigation at HKUST(GZ) NLP-LAB; advisor: Prof. Xuming Hu.

Henan University

Kaifeng, China

B.S. in Geographic Information Science, GPA: 3.68/4.0, Rank: 1/46

Sep. 2020 – Jun. 2024

- Relevant coursework: Programming, Data Structures, Database Systems, Probability and Statistics.

Research Interests

My research focuses on reliable AI for complex real-world domains, especially LLM/agent evaluation, medical knowledge updating, biosecurity risk assessment, and domain-grounded benchmark construction.

Publications and Preprints

Equal contribution is denoted by *.

1. **Dense Clinical Contrasts Enhance Medical Knowledge Updating in Large Language Models**
Yangmin Huang*, Shu Quan*, He Geng, Xin Ye, Qianyun Du, Zhiyang He, Jiaxue Hu, Xiaodong Tao
Under review, 2026
2. **A Blind Spot in Alignment: Quantifying Biosecurity Risks in Large Language Models**
Shu Quan*, Tianfang Hao*, Sitong Fang, He Geng, Jiayi Zhou, Boyuan Chen, Kaile Wang, Donghai Hong, Juntao Dai, Yaodong Yang, Jiaming Ji
Under review, 2026
3. **SafeCLI: Exploring System-Centric Risks in CLI Agents via Augmented Observability**
Donghai Hong*, Changhao Ju*, Shu Quan*, Yuyang Ren*, Boyuan Chen, Jiayi Zhou, Kaile Wang, Juntao Dai, Yaodong Yang, Jiaming Ji
Under review, 2026
4. **SLM-MATRIX: A Multi-Agent Trajectory Reasoning and Verification Framework for Enhancing Language Models in Materials Data Extraction**
Xin Li*, Zhixuan Huang*, Shu Quan, Cheng Peng, Xiaoming Ma
npj Computational Materials, 2025
5. **Nonlinear Effects of Blue-Green Space Variables on Urban Cold Islands in Zhengzhou Analyzed with Random Forest Regression**
Shu Quan*, Maojuan Li*, Tianqi Li, Haodong Liu, Yaohui Cui, Miaohan Liu
Frontiers in Ecology and Evolution, 2023
6. **Analysis of the PM_{2.5}-O₃ Pollution Characteristics and Its Potential Sources in Major Cities in the Central Plains Urban Agglomeration from 2014 to 2020**
Shu Quan, Miaohan Liu, Boxuan Chen, Yuehua Huang, Meijuan Wang, Qingxia Ma, Yan Han
Atmosphere, 2023

Research Experience

Peking University, PKU-Alignment Group
Research Assistant; advisor: Prof. Yaodong Yang

Sep. 2024 – Present
Beijing, China

- Built evaluation benchmarks and audits for medical knowledge updating, biosecurity risk, and CLI-agent safety, resulting in SEER-Bench, SPIKE-Bench, and SafeCLI.
- Analyzed failures missed by answer-level metrics, including outdated medical knowledge, functional biological risk, and system-level harms in CLI agents.

Alibaba Health
Research Intern, Medical LLM Agents

Apr. 2026 – Jun. 2026
Beijing, China

- Studied tool-selection and grounding failures in medical LLM agents, including guideline lookup, drug-interaction checking, and triage routing.

OPPO AI Center, Large Model Algorithm Department
Research Intern

Jun. 2025 – Oct. 2025
Shenzhen, China

- Built evaluation protocols for long-form report generation, covering planning quality, numerical claims, citation accuracy, source matching, and code-execution checks.

iFLYTEK, South China Research Institute
Research Intern

Feb. 2025 – May 2025
Guangzhou, China

- Designed rubric-based supervision for medical LLM alignment, separating correctness, quality bonuses, and safety vetoes to reduce hallucination and reward hacking.

Henan University
Undergraduate Researcher, Geospatial and Environmental Modeling

2022 – 2023
Kaifeng, China

- Conducted remote-sensing and statistical analyses of urban cold islands and atmospheric pollution, leading to first/co-first authored environmental science publications.

Honors and Awards

• Kaggle Silver Medal, LLM Adversarial Attack Competition, Rank 76/1,692 (Top 5%)	2025
• Outstanding Graduate, Henan Province	2024
• National Scholarship, China (Top 1%)	2023

Technical Skills

Programming: Python, PyTorch, Hugging Face Transformers, vLLM, LangGraph, Linux, Git
LLM methods: SFT, LoRA, reward modeling, GRPO, benchmark construction, LLM-as-judge evaluation, multi-agent orchestration
Evaluation domains: medical LLMs, biosecurity pipelines, CLI-agent safety, source-grounded factuality checking